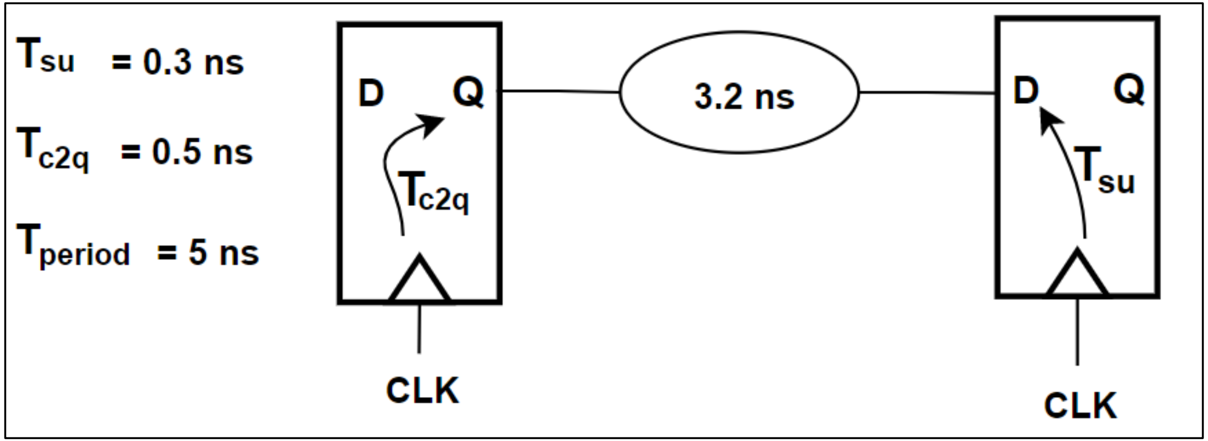
Example 1

- **Assumptions**

1. **Interconnect Delays not considered**
2. **Clock Uncertainty not considered**
3. **One Operating Condition**

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



Solution

Setup Analysis

$$Data\ Arrival\ Time = T_{cq} + T_{pd} = 0.5 + (3.2) = 3.7$$

$$Data\ Required\ Time = T_{period} - T_{su} = 5 - 0.3 = 4.7$$

$$Data\ Required\ Time > Data\ Arrival\ Time\ .$$

No Violation

$$Slack = Data\ Required\ Time - Data\ Arrival\ Time = 4.7 - 3.7 = 1$$

Example 2

- **Assumptions**

1. **Interconnect Delays is considered**
2. **Clock Uncertainty not considered**
3. **One Operating Condition**

Required: -

1. Calculate Tpd1
2. Calculate Tpd2
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths

$T_{inverter} = 0.8$ ns

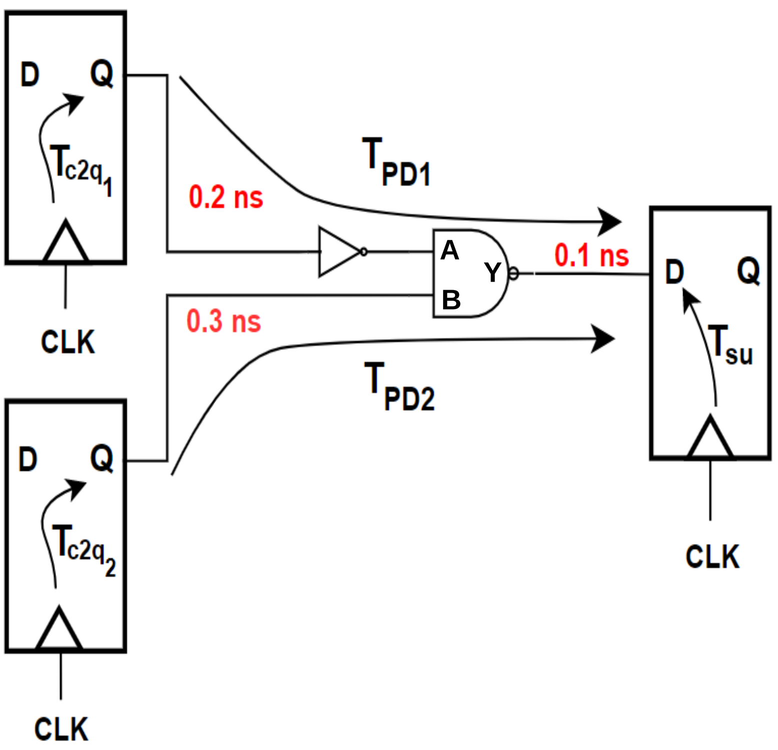
$T_{nand\ (A>Y)} = 1.75$ ns

$T_{nand\ (B>Y)} = 2.1$ ns

$T_{c2q_1} = 0.4$ ns

$T_{c2q_2} = 0.25$ ns

$T_{su} = 0.35$ ns



Solution

Setup Analysis

PATH 1

$$Data\ Arrival\ Time = T_{cq1} + T_{pd1} = 0.4 + (0.2 + 0.8 + 1.75 + 0.1) = 3.25$$

PATH 2

$$Data\ Arrival\ Time = T_{cq2} + T_{pd2} = 0.25 + (0.3 + 2.1 + 0.1) = 2.75$$

$$PATH1 > PATH2$$

$$T_{period} \geq T_{cq1} + T_{pd1} + T_{su} = 0.4 + (0.2 + 0.8 + 1.75 + 0.1) + 0.35 = 3.6$$

$$Minmum\ Clock\ period = 3.6$$

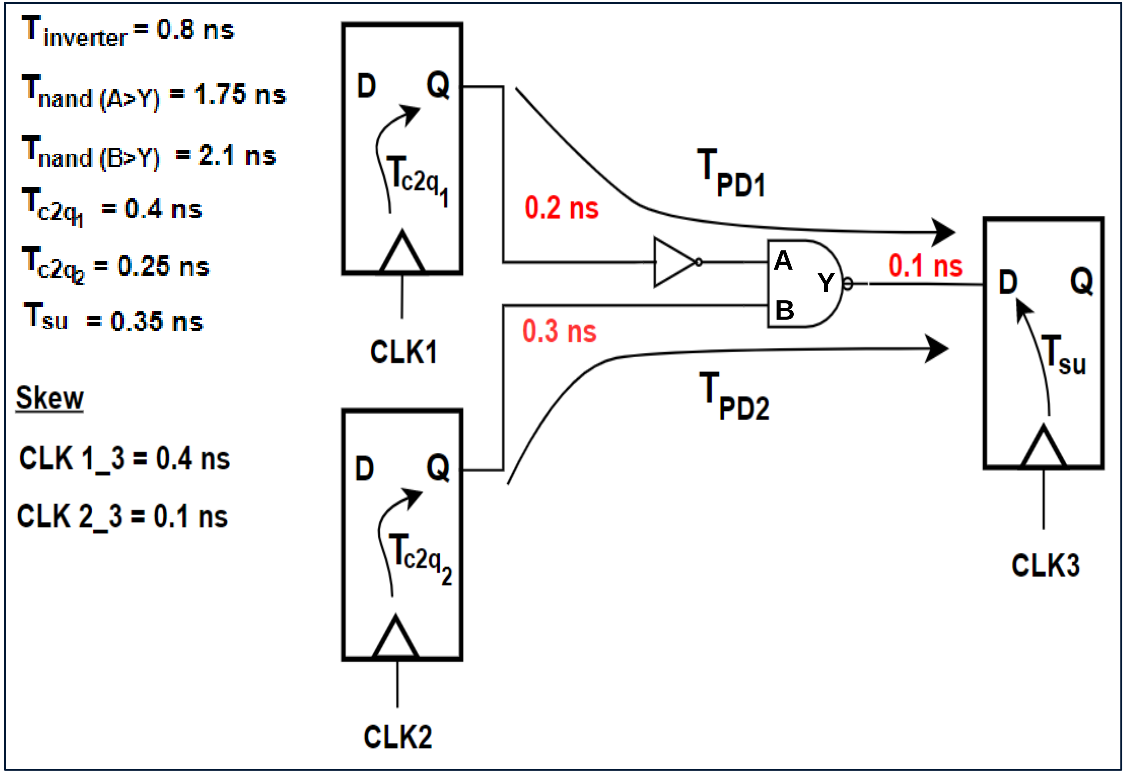
Example 3

- **Assumptions**

1. Interconnect Delays is considered
2. Clock Uncertainty is considered
3. One Operating Condition

Required: -

1. Calculate Tpd1
2. Calculate Tpd2
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths



Solution

Setup Analysis

PATH 1

$$Data\ Arrival\ Time = T_{cq1} + T_{pd1} + T_{skew\_1\_3} = 0.4 + (0.2 + 0.8 + 1.75 + 0.1) + 0.4 = 3.65$$

PATH 2

$$Data\ Arrival\ Time = T_{cq2} + T_{pd2} + T_{skew\_2\_3} = 0.25 + (0.3 + 2.1 + 0.1) + 0.1 = 2.85$$

$$PATH1 > PATH2$$

$$T_{period} \geq T_{cq1} + T_{pd1} + T_{su} + T_{skew_{13}} = 0.4 + (0.2 + 0.8 + 1.75 + 0.1) + 0.35 + 0.4 = 4$$

$$Minmum\ Clock\ period = 4$$

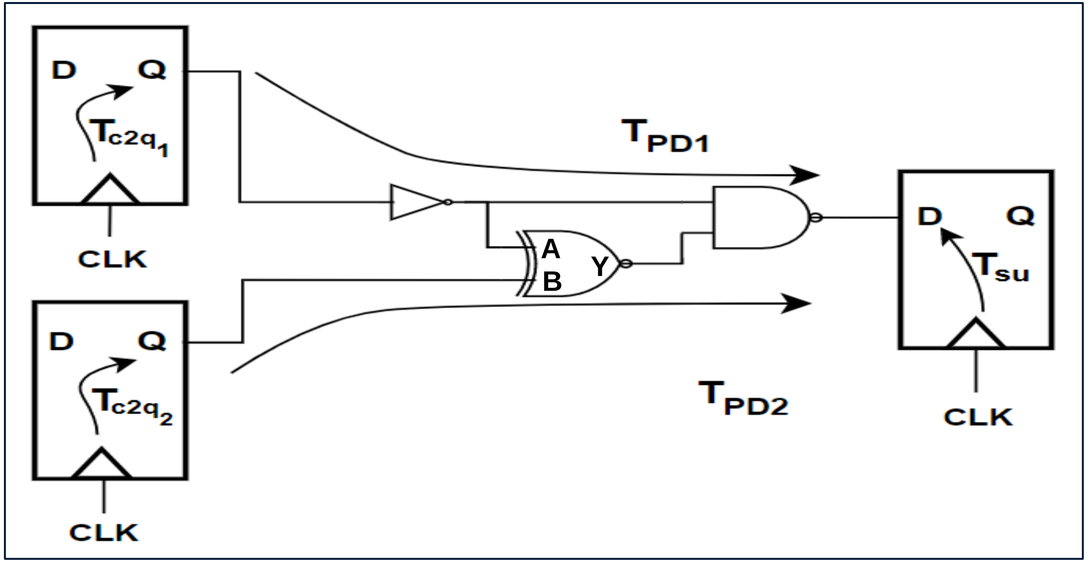
Example 4

- **Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

Required: -

1. Calculate Tpd1
2. Calculate Tpd2
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths



Delay	Inverter	XNOR	NAND	Tcq1	Tcq2	Tsu
Max	0.8	1.7	1.2	0.4	0.25	0.3
Min	0.65	1.55	1.1	0.3	0.2	0.2

Solution

Setup Analysis

PATH 1

$$Data\ Arrival\ Time = T_{cq1(max)} + T_{pd1(max)} = 0.4 + (0.8 + 1.2) = 2.4$$

PATH 2

$$Data\ Arrival\ Time = T_{cq1(max)} + T_{pd11(max)} = 0.4 + (0.8 + 1.7 + 1.2) = 4.1$$

PATH 3

$$Data\ Arrival\ Time = T_{cq2(max)} + T_{pd2(max)} = 0.25 + (1.7 + 1.2) = 3.15$$

$$PATH2\ is\ the\ critical\ Path$$

$$T_{period} \geq T_{cq1(max)} + T_{pd1(max)} + T_{su(max)} = 0.4 + (0.8 + 1.7 + 1.2) + 0.3$$

$$Minmum\ Clock\ period = 4.4$$